CN

What is a computer network?
--> A computer network is a group of two or more computers or devices connected to each other to share data, information, and resources such as files, printers, and an internet connection.
Example: Computers connected in a college computer lab form a computer network.

What are the different types of computer networks?
--> PAN (Personal Area Network), LAN (Local Area Network), MAN (Metropolitan Area Network), WAN (Wide Area Network)

What is the OSI model? --> The OSI (Open Systems Interconnection) model is a reference model that explains how data travels from one computer to another over a network. It consists of seven layers, and each layer performs a specific task in the communication process.

What are the seven layers of the OSI model?
--> Application → Presentation → Session → Transport → Network → Data Link → Physical.

What is the TCP/IP model? --> The TCP/IP (Transmission Control Protocol/Internet Protocol) model is a networking model that explains how data is transmitted between devices over a network, especially the Internet.
4 layers --> Application Layer, Transport Layer, Internet Layer, Network Access Layer

What is the difference between the OSI and TCP/IP models?
What is an IP address? --> An IP (Internet Protocol) address is a unique logical address assigned to a device on a network. It is used to identify the device and help data reach the correct destination.
There are two main versions of IP addresses: IPv4(Uses a 32-bit address) and IPv6(Uses a 128-bit address)

What is the difference between IPv4 and IPv6? --> IPv4 has a limited number of addresses, while IPv6 was introduced to provide a much larger address space as the number of Internet-connected devices increased.

What is a MAC address? --> A MAC (Media Access Control) address is a hardware address associated with a network interface. It is used to identify a device's network interface within a local network.

A MAC address is usually 48 bits long and written in hexadecimal format.

What is the difference between a MAC address and an IP address?
What is a port number?

What is a socket? --> A socket is an endpoint for communication between two devices or applications over a network. It allows them to send and receive data.
Example: In a chat application, sockets allow two users to exchange messages in real time.

What is the difference between TCP and UDP?
--> TCP (Transmission Control Protocol) is a connection-oriented transport-layer protocol that provides reliable data transmission between devices. It ensures that data arrives in the correct order and retransmits lost data when necessary.[Web browsing]
--> UDP (User Datagram Protocol) is a connectionless transport-layer protocol that sends data without establishing a connection first. It is faster and has less overhead, but it does not guarantee data delivery or correct order.[Online gaming]

Why is TCP called a connection-oriented protocol? --> TCP is called a connection-oriented protocol because it establishes a connection between the sender and receiver before transmitting data.

What is the TCP three-way handshake? --> The TCP three-way handshake is a process used to establish a connection between a client and a server before data transmission begins.
It consists of three steps: SYN[Synchronize], SYN-ACK, and ACK.

What happens during the TCP three-way handshake?
SYN: The client sends a SYN packet to request a connection with the server.
SYN-ACK: The server responds with a SYN-ACK packet to acknowledge the request and indicate that it is ready to connect.
ACK: The client sends an ACK packet to acknowledge the server's response.

What is TCP connection termination? --> TCP connection termination is the process of closing an established TCP connection after communication is complete.
It usually uses a four-way handshake involving FIN[Finish] and ACK packets

What is flow control? --> Flow control is a mechanism that prevents a sender from sending data faster than the receiver can handle.

What is congestion control? --> Congestion control is a mechanism that prevents too much data from entering the network and overloading it.

What is the difference between flow control and congestion control?

What is HTTP? --> HTTP (Hypertext Transfer Protocol) is a protocol used for communication between a web browser and a web server. It allows the browser to request web pages and the server to send responses.

What is HTTPS? --> HTTPS (Hypertext Transfer Protocol Secure) is the secure version of HTTP. It uses TLS encryption to protect data exchanged between the browser and the server.

What is the difference between HTTP and HTTPS?

What happens when you enter a URL in a browser? --> When we enter a URL in a browser, the browser first parses the URL to identify the domain name. Then, DNS resolution converts the domain name into an IP address to locate the server. Next, the browser establishes a connection with the server using TCP or QUIC. If the website uses HTTPS, a secure connection is established using TLS encryption. After that, the browser sends an HTTP request to the server. The server processes the request and sends back an HTTP response containing data such as HTML, CSS, and JavaScript files. Finally, the browser processes these files and displays the webpage to the user.

What is DNS? --> DNS (Domain Name System) translates human-readable domain names into IP addresses so that browsers can locate the correct servers.

How does DNS resolution work? --> DNS resolution is the process of converting a domain name into an IP address so that the browser can locate the correct server. When we enter a domain name like www.google.com, the browser first checks its DNS cache. If the IP address is not found, a DNS resolver looks up the address by contacting other DNS servers if necessary. Finally, the resolver returns the IP address to the browser, which uses it to connect to the server.

What is DHCP? --> DHCP (Dynamic Host Configuration Protocol) is a network protocol that automatically assigns IP addresses and other network settings, such as the default gateway and DNS server, to devices on a network. For example, when we connect our phone or laptop to Wi-Fi, DHCP can automatically assign an IP address to the device without requiring manual configuration.

What is ARP? --> ARP (Address Resolution Protocol) is used in IPv4 networks to find the MAC address associated with a known IP address on the local network. For example, when a computer wants to send data to another device on the same local network, it uses ARP to find the destination device's MAC address.

What is routing? --> Routing is the process of selecting a path for data packets to travel from a source network to a destination network. Routers use routing tables to determine where to forward packets so that they can reach the correct destination. For example, when we open a website, routing helps data packets travel from our computer through different networks to the website's server.

What is a router? --> A router is a networking device that connects two or more networks and forwards data packets between them using IP addresses. It uses a routing table to decide where to send the packets. For example, a home Wi-Fi router connects our home network to the Internet and allows multiple devices, such as phones and laptops, to communicate with external servers.

What is a switch? --> A switch is a networking device that connects multiple devices within a local area network (LAN). It uses MAC addresses to forward data to the correct destination device.

What is the difference between a router and a switch?-->[connect diff networks and connect diff networks]

What is a hub? --> A hub is a basic networking device that connects multiple devices in a LAN. Sends data to all connected ports.

What is the difference between a hub, switch, and router?

What is a firewall? --> A firewall is a security system that monitors and controls incoming and outgoing network traffic based on predefined security rules. It helps protect a computer or network from unauthorized access.

What is NAT? --> NAT (Network Address Translation) is a technique that translates IP addresses, usually between private and public addresses, allowing multiple devices on a private network to share a public IP address when accessing the Internet.

What is a proxy server? --> A proxy server acts as an intermediary between a client and another server. It forwards requests on behalf of the client and can provide features such as caching, filtering, and hiding the client's IP address from the destination server.

What is a VPN? --> A VPN (Virtual Private Network) creates an encrypted connection between your device and a VPN server over a network such as the Internet. It helps protect data in transit and can hide your public IP address from the websites you visit. Example: Using a VPN on public Wi-Fi to help protect your Internet traffic.

What is a subnet mask? --> A subnet mask is a 32-bit number used in IPv4 networks to identify which part of an IP address represents the network and which part represents the host.

What is subnetting? --> Subnetting is the process of dividing a large network into smaller networks called subnets. It helps use IP addresses efficiently and makes network management easier.

What is the difference between public and private IP addresses?
--> A public IP address is an IP address used to identify a device or network on the Internet. It must be globally unique when assigned for Internet routing and is usually provided by an Internet Service Provider (ISP).[8.8.8.8]
--> A private IP address is an IP address used within a local network, such as a home, college, or office network. It is not directly routable over the public Internet and can be reused in different private networks.[192.168.1.10]

What is localhost? --> Localhost refers to the computer you are currently using. It is used to access services running on your own machine.

What is the difference between 127.0.0.1 and 0.0.0.0? --> 127.0.0.1 means your own computer; 0.0.0.0 when binding a server means all IPv4 interfaces.

What are HTTP methods? --> HTTP methods specify the action a client wants a server to perform on a resource. [GET, POST, PUT, PATCH, and DELETE]

What is the difference between GET, POST, PUT, PATCH(Partially), and DELETE?

What are HTTP status codes?
What is the difference between 401 and 403 status codes?
What is the difference between 301 and 302 status codes?
What are cookies and sessions?
--> Cookies are small pieces of data stored in the user's browser. They can be used to remember preferences, maintain login state, or store session identifiers.

--> Sessions are used to maintain user-specific state across multiple requests. In a traditional server-side session system, the server stores session data and the browser stores a session ID, often in a cookie.

Example: After logging in, a session can help keep you logged in while navigating between pages.

What is the difference between WebSocket and HTTP? --> HTTP follows a request-response model, where the client sends a request to the server and the server sends back a response. It is commonly used for loading web pages and communicating with REST APIs. In contrast, WebSocket provides full-duplex, two-way communication over a persistent connection, allowing both the client and server to send data at any time without repeatedly establishing new requests. HTTP is commonly used to fetch user profiles or product details, whereas WebSocket is used in real-time applications such as chat apps, online gaming, and live notifications.

DBMS

1. What is a DBMS?
--> “DBMS is basically a software that helps us store and manage data in a structured way. It allows us to easily insert, update, delete, and retrieve data, while also handling things like security and data consistency. MySQL and PostgreSQL are some common examples.”

2. what is RDBMS?
--> “RDBMS stands for Relational Database Management System. It is a type of DBMS where data is stored in the form of tables, with rows and columns. These tables can also be related to each other using keys like primary keys and foreign keys. MySQL, PostgreSQL, and Oracle are examples of RDBMS.”

3. What is the difference between DBMS and RDBMS?
--> “The main difference is that RDBMS is a type of DBMS. A DBMS can store data without necessarily having relationships between different data sets, whereas an RDBMS stores data in tables and allows those tables to be related using keys like primary key and foreign key. RDBMS also generally provides stronger support for data integrity and consistency. For example, MySQL and PostgreSQL are RDBMS.”

4. What is a database?
--> “A database is basically a structured collection of data where we can store, manage, and retrieve information easily. For example, in a college database, we can store student details like their name, ID, department, and marks.”

5. What is a relational database?
--> “A relational database is a type of database where data is stored in tables, and these tables can be connected to each other using relationships. For example, we can have a Student table and a Department table, and connect them using a common key.”

6. What is a table?
--> “A table is basically a way of organizing data in a database using rows and columns. Each row represents one record, and each column represents a particular attribute of that record. For example, in a Student table, one row can represent one student, while columns can contain their name, ID, department, and marks.”

7. What is a tuple and an attribute?
--> “A tuple is basically a single row or record in a table, while an attribute is a column that represents a particular property of the data. For example, in a Student table, one student's complete information is a tuple, and Name, ID, and Department are attributes.”

8. What is a Primary Key?
--> “A primary key is a column, or a combination of columns, that uniquely identifies each record in a table. It cannot contain duplicate or NULL values. For example, Student_ID can be a primary key in a Student table.”

9. What is a Foreign Key?
--> “A foreign key is a column that is used to create a relationship between two tables. It usually refers to the primary key of another table. For example, Department_ID in a Student table can refer to Department_ID in a Department table.”

10. What is a Candidate Key?
--> “A candidate key is a column, or a combination of columns, that can uniquely identify each record and is eligible to become a primary key. There can be multiple candidate keys in a table, but we choose one of them as the primary key.
For example, in a Student table, both Student_ID and Email can uniquely identify a student. So, both can be candidate keys. If we choose Student_ID as the primary key, then Email becomes an alternate key.”

11. What is a Super Key?
--> “A super key is a column or combination of columns that can uniquely identify a record in a table. It can contain extra columns that are not required for uniqueness.
For example, in a Student table, Student_ID alone can uniquely identify a student, so it is a super key. Student_ID + Name can also uniquely identify the student, so that is also a super key, but Name is an extra column here. So, every candidate key is a super key, but every super key is not necessarily a candidate key.”

12. What is an Alternate Key?
--> “An alternate key is a candidate key that was not selected as the primary key. For example, if both Student_ID and Email can uniquely identify a student and we choose Student_ID as the primary key, then Email becomes an alternate key.”

13. What is a Composite Key?
--> “A composite key is a key made up of two or more columns that together uniquely identify a record. For example, in an Enrollment table, Student_ID and Course_ID together can form a composite key.”

14. What is the difference between a primary key and a unique key?
--> “Both primary key and unique key are used to ensure that values are not duplicated. The main difference is that a table can have only one primary key, while it can have

multiple unique keys. A primary key also does not allow NULL values, whereas a unique key can generally allow NULL values, depending on the database system.
For example, in a Student table, Student_ID can be the primary key, while Email can be a unique key to make sure two students don't have the same email.”

15. What is normalization?
--> “Normalization is basically a process of organizing data in a database to reduce duplicate data and avoid problems when we insert, update, or delete data. We usually divide a large table into smaller related tables and connect them using keys. For example, instead of storing student and department information repeatedly in one table, we can create separate Student and Department tables.”

16. What are 1NF, 2NF, and 3NF?
-->
1NF: Each column should contain atomic values, meaning no multiple values in a single cell.
2NF: The table should be in 1NF, and every non-key column should depend on the entire primary key, not just part of it.
3NF: The table should be in 2NF, and non-key columns should depend only on the primary key, not on another non-key column.
2NF checks dependency on the complete key where_as 3NF checks dependency on another non-key column

17. What is BCNF?
--> “BCNF stands for Boyce-Codd Normal Form. It is a stronger version of 3NF. A table is in BCNF if, for every functional dependency, the determinant is a candidate key.”

18. What is Denormalization?
--> “Denormalization is the process of adding duplicate data to reduce joins and improve database read performance.”

19. What is a functional dependency?
--> A functional dependency means one column determines the value of another column in a table.[Example: Student_ID → Student_Name.]

20. What is a partial dependency? --> A partial dependency occurs when a non-key column depends on only a part of a composite primary key, instead of the entire key.

22. What is a transitive dependency? --> A transitive dependency occurs when a non-key column depends on another non-key column instead of directly depending on the primary key.

23. What are insertion, update, and deletion anomalies?
--> Insertion anomaly: Difficulty inserting new data without other unnecessary data.
Update anomaly: The same information must be updated in multiple places, which can cause inconsistency.
Deletion anomaly: Deleting one record accidentally removes other important information.

24. What is SQL? --> SQL (Structured Query Language) is a language used to create, retrieve, update, and delete data in relational databases.

25. What are DDL, DML, DCL, and TCL? -->
DDL (Data Definition Language): Used to define database structures. Example: CREATE, ALTER, DROP.
DML (Data Manipulation Language): Used to modify data. Example: INSERT, UPDATE, DELETE.
DCL (Data Control Language): Used to control database permissions. Example: GRANT, REVOKE.
TCL (Transaction Control Language): Used to manage transactions. Example: COMMIT, ROLLBACK, SAVEPOINT.

26. What is the difference between DELETE, DROP, and TRUNCATE?
--> Removes selected rows or all rows from a table, Removes the entire table, including its structure, Removes all rows from a table but keeps its structure.

27. What is the difference between WHERE and HAVING(GROUP BY)?

28. What is the difference between GROUP BY and ORDER BY? -->
- Groups rows with the same values
- Sorts the result in ascending or descending order.

29. What are aggregate functions in SQL?
--> Aggregate functions perform calculations[COUNT(), SUM(), AVG(), MIN(), and MAX()]

30. What are joins in SQL?
--> Joins are used to combine data from two or more tables based on a related column.

31. What is the difference between INNER JOIN, LEFT JOIN, RIGHT JOIN, and FULL JOIN? -->
INNER JOIN: Returns matching rows from both tables.
LEFT JOIN: Returns all rows from the left table and matching rows from the right table.
RIGHT JOIN: Returns all rows from the right table and matching rows from the left table.
FULL OUTER JOIN: Returns matching and non-matching rows from both tables.

32. What is a self join? --> A self join is a join in which a table is joined with itself to compare rows within the same table.

33. What is a subquery?
--> A subquery is a query written inside another SQL query.
Example: Finding employees whose salary is greater than the average salary.

34. What is the difference between a subquery and a join?
35. What is a correlated subquery?
36. What is a view? --> A view is a virtual table created using an SQL query. It displays data from one or more tables without necessarily storing the data itself.

37. What is an index? --> An index is a database structure that helps find data in a table more quickly.

38. Why are indexes used? --> Indexes are used to speed up data retrieval and make searching, filtering, and joining tables faster.

39. What are the advantages and disadvantages of indexes?
40. What is a clustered index and a non-clustered index?
41. What is a transaction? --> A transaction is a group of database operations performed as a single unit. Either all operations are completed successfully, or the changes are rolled back.
Example: Transferring money from one bank account to another.

42. What are ACID properties? -->
Atomicity: All operations succeed, or none do.
Consistency: The database remains valid before and after a transaction.
Isolation: Transactions do not interfere with each other in an incorrect way.
Durability: Once committed, changes are permanently saved.

43. What is concurrency control? --> Concurrency control is a technique used to manage multiple transactions running at the same time without causing incorrect data or inconsistencies.

44. What are dirty reads, non-repeatable reads, and phantom reads? -->
Dirty read: Reading data that another transaction has not committed yet.
Non-repeatable read: Reading the same row twice and getting different values because another transaction updated it.
Phantom read: Running the same query twice and getting different sets of rows because another transaction inserted or deleted matching rows.

45. What is a deadlock in DBMS? --> A deadlock occurs when two or more transactions wait for resources held by each other, so none of them can proceed.

46. What is database isolation level? --> An isolation level determines how much one transaction is protected from changes made by other concurrent transactions.

47. What is the difference between optimistic and pessimistic locking? -->
Optimistic locking: Assumes conflicts are rare and checks for conflicts when updating data.
Pessimistic locking: Locks data to prevent conflicting operations while a transaction is using it.

48. What is a stored procedure? --> A stored procedure is a set of SQL statements stored in the database that can be executed whenever needed.

49. What is a trigger? --> A trigger is a database program that automatically executes when a specified event, such as INSERT, UPDATE, or DELETE, occurs on a table.

50. What is the difference between a stored procedure and a function? -->
Stored procedure: Performs a task and can return results through output parameters or result sets.
Function: Typically calculates and returns a value and can often be used inside SQL expressions.